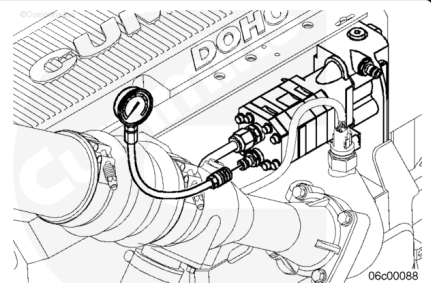


Trainings ▾

Initial Check

With Pressurized Fuel Filtering

Install a pressure gauge, Part Number 3824877 (2758 kPa [400 psi]), on the fuel pump Compuchek® fitting to measure the fuel pressure before the fuel filter.

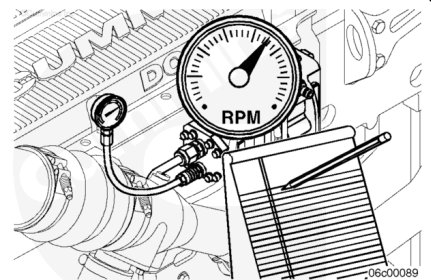


Start the engine and run at high idle and no load.

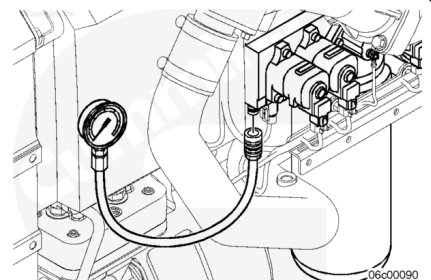
Measure and record the fuel pressure.

Shut the engine off.

Remove the pressure gauge from the fuel pump.



Install a pressure gauge, Part Number 3824877 (275 kPa [400 psi]), on the rail fuel pressure Compuchek® fitting located on the integrated fuel system module to measure fuel pressure after the filter.



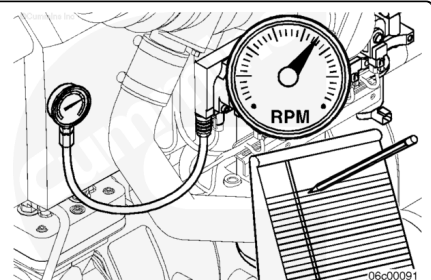
Start the engine and operate at high idle and no load.

Measure and record the fuel pressure.

Shut the engine off.

Remove the pressure gauge from the rail fuel pressure Compuchek® fitting.

The pressure difference between the rail fuel pressure and fuel pump pressure is the differential pressure.



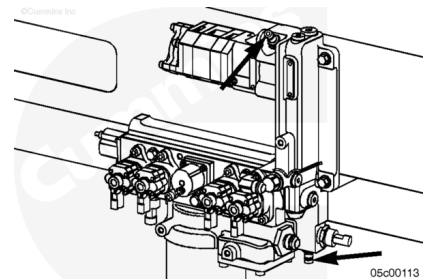
kpa		psi
517	MAX	75

If the fuel filter restriction is above specifications, replace the filter.

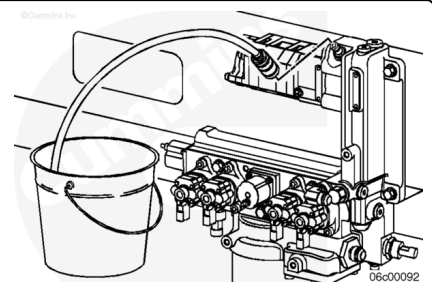
With Vacuum Fuel Filtering

Connect a vacuum gauge to either suction-side Compuchek® fitting.

Note : Some engines equipped with priming pumps do not have a lower Compuchek® fitting and have an air vent line plumbed into the upper location. Refer to the following procedure for measuring inlet restriction.



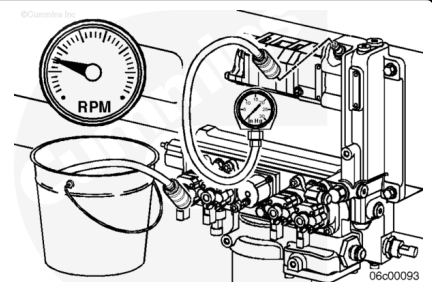
Disconnect the air bleed line from the upper location on the IFSM. Insert a Compuchek® fitting in this location and run a line to a catch container.



Start the engine.

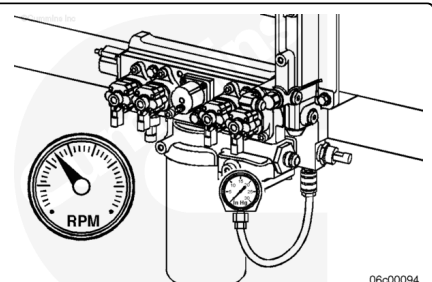
After the fuel lift pump stops running and with the engine operating at idle, disconnect the hose to the catch container.

Connect the vacuum gauge to the Compuchek® fitting and measure the fuel inlet restriction.



Operate the engine at high idle and no load.

	mm-hg		in-hg	
Upper Location		356	MAX	14
Lower Location		305	MAX	12



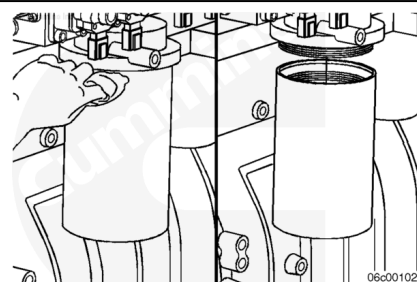
An inlet restriction greater than specification indicates either a dirty fuel filter or a restriction in the OEM fuel supply plumbing. Inlet restriction at the fuel supply connection should be checked to identify the correct source of the restriction.

Remove

With Pressurized Fuel Filtering

⚠ WARNING ⚠

Fuel is flammable. Keep all cigarettes, flames, pilot lights, arcing equipment, and switches out of the work area and areas sharing ventilation to reduce the possibility of severe personal injury or death when working on the fuel system.



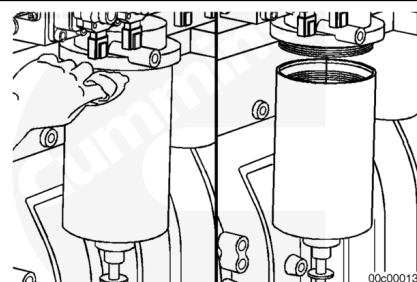
Clean the area around the fuel filter head and filter.

Remove the fuel filter with filter wrench, Part Number 3375049.

With Vacuum Fuel Filtering

⚠ WARNING ⚠

Fuel is flammable. Keep all cigarettes, flames, pilot lights, arcing equipment, and switches out of the work area and areas sharing ventilation to reduce the possibility of severe personal injury or death when working on the fuel system.



Clean the area around the fuel filter head and filter.

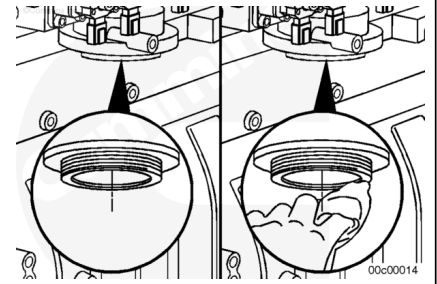
Disconnect the wiring harness from the water-in-fuel sensor if equipped.

Remove the fuel filter with filter wrench, Part Number 3376807.

Clean and Inspect for Reuse

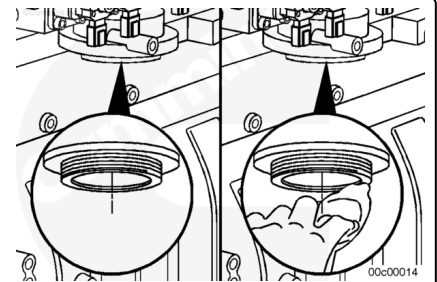
With Pressurized Fuel Filtering

Use a clean, lint-free cloth to clean the filter head gasket surface.



With Vacuum Fuel Filtering

Use a clean, lint-free cloth to clean the filter head gasket surface.



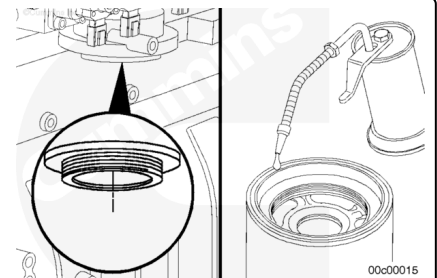
Install

With Pressurized Fuel Filtering

Use the correct filter(s) for your engine. It **must** remove a minimum of 95 percent of free and emulsified water. It **must** also have a minimum of 98.7 percent 15-micron particle-removal efficiency.

Cummins, Part Number 3101546

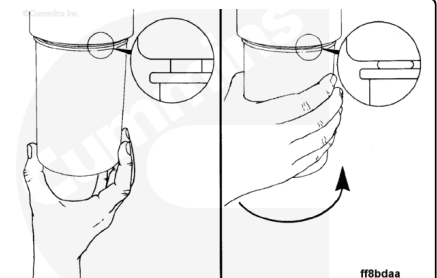
Fleetguard® Nelson®, Part Number FF2200



Apply a thin coating of clean engine oil to the filter gasket surface.

⚠ CAUTION ⚠

Mechanical overtightening of the filter can distort the threads or damage the filter element seal.



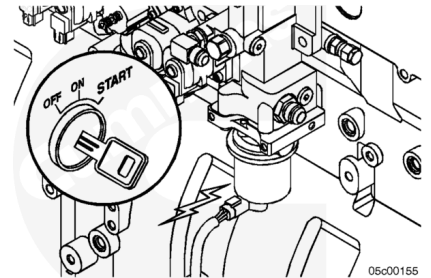
Note : Engines equipped with priming pumps do not require the fuel filter to be filled prior to installation if the following filling procedure is followed.

Install the filter onto the filter head. Turn the filter until the gasket contacts the filter head surface.

Tighten the filter an additional 3/4 of a turn after the gasket contacts the filter head surface, or as specified by the filter manufacturer.

Fill the fuel filter by turning the ignition switch to the ON position. The priming pump will run for two minutes which will adequately fill the fuel filter. The engine can then be started.

Note : The engine will, perhaps, run rough for several minutes until the air is out of the system.



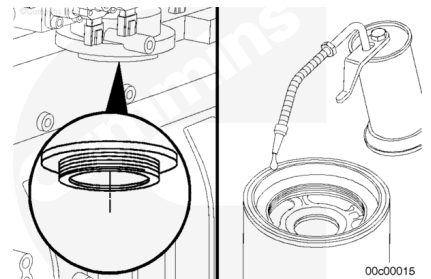
With Priming Pump

Use the correct filter(s) for your engine. It **must** remove a minimum of 95 percent of free and emulsified water. It **must** also have a minimum of 98.7 percent at 25-micron particle-removal efficiency.

Cummins, Part Number 4010651

Fleetguard® Nelson®, Part Number FS1040

Apply a thin coating of clean engine oil to the filter gasket surface and the center seal.



⚠ CAUTION ⚠

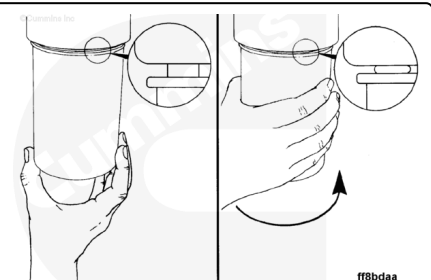
Mechanical overtightening of the filter can distort the threads or damage the filter element seal.

Note : Engines equipped with priming pumps do not require the fuel filter to be filled prior to installation if the following filling procedure is followed.

Install the filter onto the filter head. Turn the filter until the gasket contacts the filter head surface.

Tighten the filter an additional 3/4 of a turn after the gasket contacts the filter head surface, or as specified by the filter manufacturer.

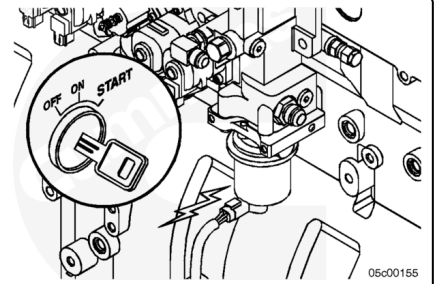
Note : If the filter is equipped with a water-in-fuel sensor, rotate the sensor to the desired location and connect the wiring harness.



Fill the fuel filter by turning the ignition switch to the ON position. The priming pump will run for two minutes which will adequately fill the filter. The engine can then be started.

Some engines utilize a manual, remote mount priming pump. Flip the priming pump mounted toggle switch to the ON position. Run the pump for two minutes and shut it off. The engine can then be started.

Note : The engine will, perhaps, run rough for several minutes until the air is out of the system.



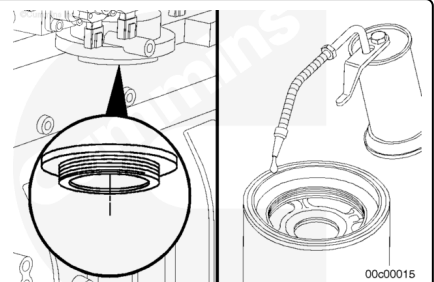
Without Priming Pump

Use the correct filter(s) for your engine. It **must** remove a minimum of 95 percent of free and emulsified water. It **must** also have a minimum of 98.7 percent at 25-micron particle-removal efficiency.

Cummins, Part Number 4010651

Fleetguard® Nelson®, Part Number FS1040

Apply a thin coating of clean engine oil to the filter gasket surface and the center seal.



⚠ CAUTION ⚠

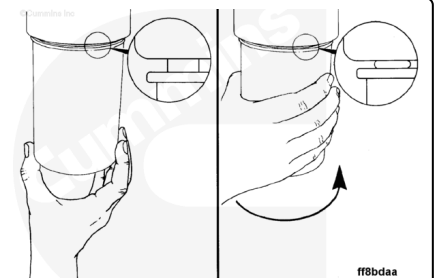
Mechanical overtightening of the filter can distort the threads or damage the filter element seal.

Note : Fill the filter with clean fuel prior to installation.

Install the filter onto the filter head. Turn the filter until the gasket contacts the filter head surface.

Tighten the filter an additional 3/4 of a turn after the gasket contacts the filter head surface, or as specified by the filter manufacturer.

Note : If the filter is equipped with a water-in-fuel sensor, rotate the sensor on the filter to the desired location and connect the wiring harness.



Remove the external hex plug on the top of the integrated fuel system module. Crank the engine until a solid stream of fuel comes out of the port.

Install the hex plug.

Crank the engine for 20 seconds. If the engine does **not** start within 20 seconds, wait two minutes. It will probably be necessary to remove the filter, fill the filter with clean fuel, and install the filter.

Repeat these steps until the engine starts.

Note : The engine will, perhaps, run rough for several minutes until the air is out of the system.

